INCH-POUND

MIL-PRF-19500/619B
28 April 2000
SUPERSEDING
MIL-PRF-19500/619A
11 November 1997

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SCHOTTKY POWER RECTIFIERS,
TYPES 1N6849 – 1N6856, 1N6849U1 – 1N6856U1
1N6849U2 – 1N6856U2 and 1N6849U3 – 1N6856U3
JAN, JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for schottky power surface mount rectifiers. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (surface mount).

1.3 Maximum ratings.

Column 1	Column 2	Column 3	Column 4	Column 5		Column 6	
Types 1/	Peak inverse voltage	I_O 2/ T_C = +100°C	I_{FSM} 3/	Maximum 4/ thermal resistance		T_{STG}	T_{OP}
	V dc	A dc	A (pk)	°C/W 5/	°C/W 6/	°C	°C
1N6849/U1/U2/U3	30	110	1,000	0.20	0.45	-65 to +175	-65 to +150
1N6850/U1/U2/U3	45	150	1,000	0.20	0.45		-65 to +175
1N6851/U1/U2/U3	100	120	900	0.20	0.45		-65 to +175
1N6852/U1/U2/U3	30	60	900	0.35	0.80		-65 to +150
1N6853/U1/U2/U3	45	75	900	0.35	0.80		-65 to +175
1N6854/U1/U2/U3	100	60	900	0.35	0.80		-65 to +175
1N6855/U1/U2/U3	45	10	300	0.85	1.40		-65 to +175
1N6856/U1/U2/U3	100	10	300	0.85	1.40		-65 to +175

1/ See 1.5 for type designation/lead configuration.
2/ Derate the I_O linearly to zero at 100°C to the maximum operating temperature.
3/ I_{FSM}, t_p = 8.3 ms, T_C = +25°C, V_R = PIV.
4/ Maximum thermal resistance junction to cathode surface.
5/ Thermal resistance value for package outline basic version, U1 version and U2 version.
6/ Thermal resistance value for package outline U3 version.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

1.4 Primary electrical characteristics at T_C = 25°C.

Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13	Column 14
Types	Maximum forward voltage per leg T_C = +25°C		Maximum forward voltage per leg T_C = +125°C		Maximum reverse current per leg at PIV mA		Maximum junction capacitance per leg 1 KHz ≤ f ≤ 1 MHz V_R = 5 V
	Volts	Amperes	Volts	Amperes	T_J = +25°C	T_J = +125°C	pF
1N6849/U1/U2/U3 1N6850/U1/U2/U3 1N6851/U1/U2/U3	0.53 0.75 0.90	110 150 120	0.43 0.65 0.73	110 150 120	7.5 5.0 5.0	400 1/ 150 100	7,500 5,000 2,700
1N6852/U1/U2/U3 1N6853/U1/U2/U3 1N6854/U1/U2/U3	0.55 0.74 0.90	60 75 60	0.49 0.70 0.83	60 75 60	5.0 5.0 1.5	200 1/ 45 20	3,700 2,600 1,400
1N6855/U1/U2/U3 1N6856/U1/U2/U3	0.55 0.75	10 10	0.49 0.63	10 10	2.0 1.0	15 10	1,000 600

1/ T_J = +110°C.

1.5 Type designation. The type designation used under this part will be as follows:

JAN, JANTX, JANTXV1N68xx/U1, /U2, /U3

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

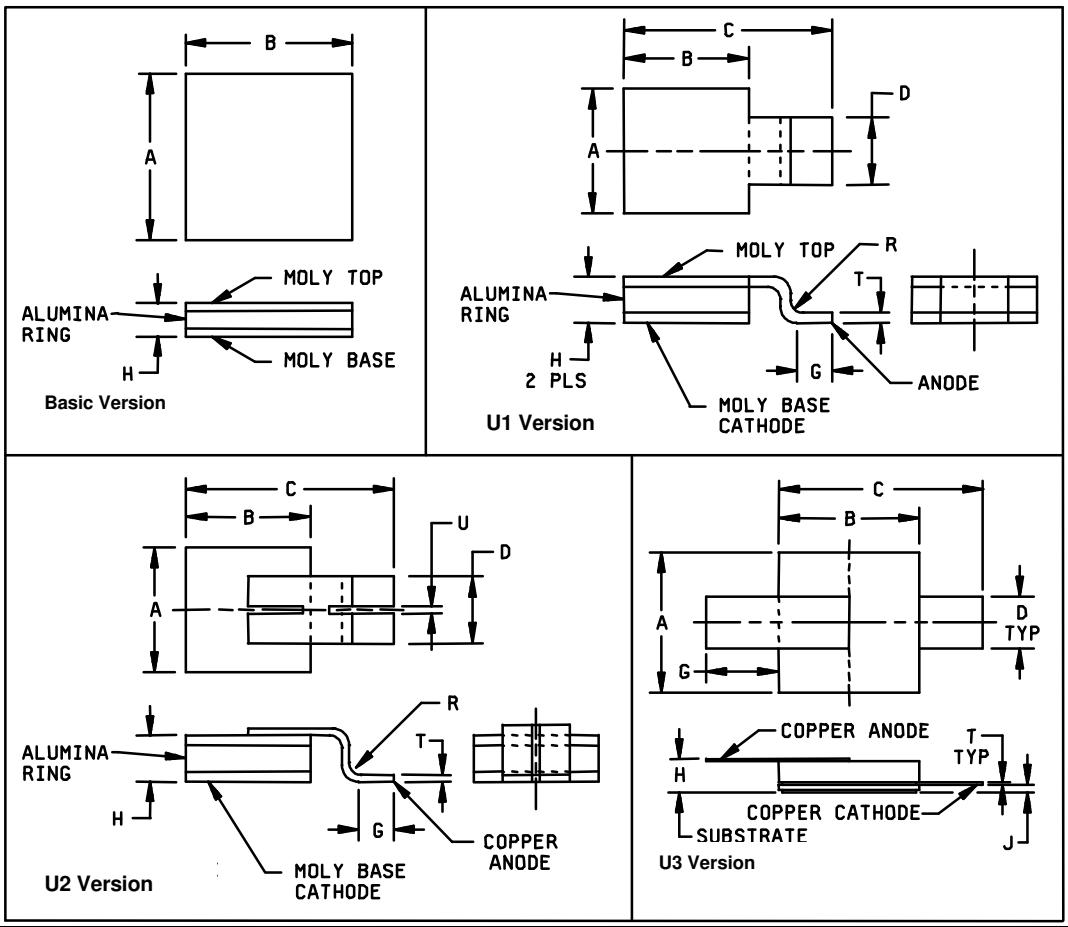
STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.



Package outline	A	B	C	D	G	H	J	R	T	U
1N6855, 1N6856	0.225	0.225	- - -	- - -	- - -	0.075 max	- - -	- - -	- - -	- - -
1N6852 - 1N6854	0.315	0.315	- - -	- - -	- - -	0.075 max	- - -	- - -	- - -	- - -
1N6849 - 1N6851	0.410	0.410	- - -	- - -	- - -	0.075 max	- - -	- - -	- - -	- - -
1N6855U1, 1N6856U1	0.225	0.225	0.375	0.125	0.060	0.075 max	- - -	0.020	0.020	- - -
1N6852U1 - 1N6854U1	0.315	0.315	0.465	0.170	0.060	0.075 max	- - -	0.020	0.020	- - -
1N6849U1 - 1N6851U1	0.410	0.410	0.550	0.200	0.060	0.075 max	- - -	0.020	0.020	- - -
1N6855U2, 1N6856U2	0.225	0.225	0.375	0.150	0.060	0.090 max	- - -	0.020	0.015	0.015
1N6852U2 - 1N6854U2	0.315	0.315	0.465	0.150	0.060	0.090 max	- - -	0.020	0.015	0.015
1N6849U2 - 1N6851U2	0.410	0.410	0.560	0.150	0.060	0.090 max	- - -	0.020	0.015	0.015
1N6855U3, 1N6856U3	0.225	0.225	0.405	0.150	0.190	0.138 max	0.037	- - -	0.012	- - -
1N6852U3 - 1N6854U3	0.315	0.315	0.495	0.150	0.190	0.138 max	0.037	- - -	0.012	- - -
1N6849U3 - 1N6851U3	0.410	0.410	0.590	0.150	0.190	0.138 max	0.037	- - -	0.012	- - -
Tolerances	±0.010	±0.010	±0.020	±0.010	±0.010	- - -	±0.005	±0.005	±0.002	±0.005

NOTE: All dimensions are in inches.

FIGURE 1. Physical dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Base, lid, and lead finish. Base and lid material shall be Molybdenum. Lead finish shall be tin-plated or solder-dipped. When a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500. In lieu of device marking all marking may be omitted from the body except for lot date code and polarity, but shall be retained on the initial container.

3.6 Electrostatic discharge protection. The devices covered by this specification shall be classified for ESD sensitivity in accordance with MIL-PRF-19500 and MIL-STD-750, method 1020, and shall be handled, marked, and packaged accordingly.

3.7 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurements JANTX and JANTXV levels
Thermal response 1/	MIL-STD-750, method 3101 (see 4.5.3)
9	Not applicable
10	MIL-STD-750, method 1038, test condition A T_C = +100°C
11	Subgroup 2 of table I herein
12	Not applicable
13	Subgroup 2 of table I herein; $\Delta I_R \leq 5.0$ mA or 100 percent of initial value, whichever is greater. $\Delta V_F \leq .05$ volts.

1/ Shall be performed anytime before screen 10.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with 4.3, screen 13.

4.4.2.1 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B1	2026	Inspect all solder covered surface
B2	2031	Prior to hemeticity, 5 cycles, 5 seconds per cycle. Electrical measurements: end points shall be in accordance with Group A Subgroup 2.
B3	1037	I_O set to maintain T_C = +100°C, ΔT_C = 85°C, +15°C, -5°C; $t_{on} = t_{off}$ = 1 minute minimum for 2,000 cycles.
B4	2075	See 3.3.2.
B5	4081	See 4.5.2.

NOTE: Subgroups 2, 3, and 6, electrical measurements, see 4.3, screen 13.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with 4.3, screen 13).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

4.4.3.1 Group C inspection, appendix E, table V of MIL-PRF-19500.

Subgroup	Method	Conditions
C3	2016	1,500 G's, 0.5 ms.
C6	1037	I_O set to maintain T_C = +100°C, ΔT_C = 85°C, +15°C, -5°C; t_{on} = t_{off} = 1 minute minimum for 6,000 cycles.

NOTE: Subgroups 2, 3, and 6, electrical measurements, see 4.3, screen 13.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions for subgroup testing in table II herein. Electrical measurements shall be in accordance with 4.3, screen 13.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurement shall be conducted in accordance with test method 4081 of MIL-STD-750. The maximum limit of $R_{\theta JC}$(max) shall be in accordance with 1.3.

4.5.3 Thermal response. The ΔV measurement shall be performed in accordance with method 3101 of MIL-STD-750. The ΔV conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves. The following parameter measurements shall apply:

a. Measuring current (I_M) ..... +100 mA.

b. Heating current (I_H)..... 20 A minimum.

c. Heating time (t_H)..... 100 ms.

d. Measurement time delay (t_{MD})..... +10 to 100 µs.

e. Sample window time (t_{SW}) ..... 5 µs maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions 2/		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	t_p = 300 µs I_F = Column 9	V_{FM}		Col 8	V (pk)
Reverse current	4016	V_R = Column 2	I_{R1}		Col 12	mA dc
Subgroup 3						
High temperature operation:		T_C = +125°C				
Reverse current	4016	V_R = Column 2	I_{R2}		Col 13	mA dc
Subgroup 4						
Capacitance (per leg)	4001	V_R = 5 V; 1 kHz ≤ f ≤ 1 MHz	C_J		Col 14	pF
Subgroup 5						
Surge current	4066	I_{FSM} = rated (Column 4), 10 surges of 8.3 ms each at 1 minute intervals; V_{RWM} = rated PIV (Column 2)				
Electrical measurements		see 4.3 screen 13.				

1/ For sampling plan, see MIL-PRF-19500.
2/ See 1.3 and 1.4 for conditions and limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sample size
	Method	Conditions	
Subgroup 1 Thermal shock Electrical measurement	1051	-55°C to +150°C, 500 cycles See 4.3, screen 13	22 c = 0
Subgroup 2 2/ Steady-state reverse bias Electrical measurement	1038	Test condition A; 1,000 hours; T_J = +150°C; V_R = 80-85 percent rated PIV. See 4.3, screen 13	22 c = 0
Subgroup 3 Not applicable			
Subgroup 4 2/ Thermal resistance, junction to lead	3101 or 4081	$R_{\theta JC}$ = Column 5 (°C/W maximum) See 4.5.2	11 c = 0
Subgroup 5 Not applicable			
Subgroup 6 ESD	1020		11 c = 0

1/ For initial design verification only (one time testing).
2/ A separate sample may be pulled for each test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-2296)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/619B	2. DOCUMENT DATE

3. DOCUMENT TITLE: SEMICONDUCTOR DEVICE, DIODE, SCHOTTKY POWER RECTIFIERS, TYPES 1N6849 - 1N6856, 1N6849U1 - 1N6856U1, 1N6849U2 - 1N6856U2 and 1N6849U3 - 1N6856U3, JAN, JANTX AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS *(Include Zip Code)* DSCC-VAC 3900 E. Broad Street Columbus, Ohio 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:14Z
Check the source to verify that this is the current version before use.